

Working with Lists, Tuples, and Dictionaries

Lab overview

In Python, string and numeric data types are often used in groups called *collections*. Three such collections that Python supports are the list, the tuple, and the dictionary.

In this lab, you will:

- Use the list data type
- Use the tuple data type
- Use the dictionary data type

Estimated completion time

45 minutes

Exercise 1: Introducing the list data type

Accessing the AWS Cloud9 IDE

1 2 3 Start your lab environment by going to the top of these instructions and choosing **Start Lab**. 17

A **Start Lab** panel opens, displaying the lab status.

2. Wait until you see the message *Lab status: ready*, and then close the **Start Lab** panel by choosing the **X**.
3. At the top of these instructions, choose **AWS**.

The AWS Management Console opens in a new browser tab. The system automatically logs you in.

Note: If a new browser tab does not open, a banner or icon at the top of your browser typically indicates that your browser is preventing the site from opening pop-up windows. Choose the banner or icon, and choose **Allow pop ups**.

4. In the AWS Management Console, choose **Services > Cloud9**. In the **Your environments** panel, locate the **reStart-python-cloud9** card, and choose **Open IDE**.

The AWS Cloud9 environment opens.

Note: If a pop-up window opens with the message *.c9/project.settings have been changed on disk*, choose **Discard** to ignore it. Likewise, if a dialog window prompts you to *Show third-party content*, choose **No** to decline.

Creating your Python exercise file

5. From the menu bar, choose **File > New From Template > Python File**.

This action creates an untitled file.

6. Delete the sample code provided from the template file.
7. Choose **File > Save As...**, and provide a suitable name for the exercise file (for example, *collections.py* and save it under the **/home/ec2-user/environment** directory.

Accessing the terminal session

8. In your AWS Cloud9 IDE, choose the **+** icon and select **New Terminal**.

A terminal session opens.

9. To display the present working directory, enter `pwd`. This command points to **/home/ec2-user/environment**.

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
 10. In this directory, you should also be able to locate the file you created in the previous section.

Defining a list

In this activity, you will edit a Python script to hold a collection of fruit names, or a list of fruit.

11. From the navigation pane of the IDE, choose the **.py** file that you created in the previous *Creating your Python exercise file* section.

12. In the file, enter the following code:

```
myFruitList = ["apple", "banana", "cherry"]  
print(myFruitList)  
print(type(myFruitList))
```

13. Save and run the file.

14. Confirm that the script runs correctly and that the output displays as you expect it to.

Accessing a list by position

You can access the contents of a list by position. In this activity, you will print out each item in our list by their position:

15. In programming languages, the list position starts at zero (0). The brackets tell Python which position in the list you want. To access the `apple` string, enter the following code:

```
print(myFruitList[0])
```

16. To access the `banana` string, enter the following:

```
print(myFruitList[1])
```

17. To access the `cherry` string, enter the following code:

```
print(myFruitList[2])
```

18. Save and run the file.

19. Confirm that the script runs correctly and that the output displays as you expect it to.

Changing the values in a list

1 The values of a list can be changed. In this activity, you will change `cherry` to `orange`. 16 17

20. In Python, list position starts at zero (0), so you must use the numeral 2 to access the third position. Enter the following code:

```
myFruitList[2] = "orange"
```

21. Print the updated list:

```
print(myFruitList)
```

22. Save and run the file.

23. Confirm that the script runs correctly and that the output displays as you expect it to.

```
['apple', 'banana', 'cherry']  
<class 'list'>  
apple  
banana  
cherry  
['apple', 'banana', 'orange']
```

Exercise 2: Introducing the tuple data type

Defining a tuple

The tuple is like a list, but it can't be changed. A data type that can't be changed after it's created is said to be *immutable*. To define a tuple, you use parentheses instead of brackets ([]).

24. Create a tuple by entering the following code:

```
myFinalAnswerTuple = ("apple", "banana", "pineapple")  
print(myFinalAnswerTuple)  
print(type(myFinalAnswerTuple))
```

25. Save and run the file.

26. Confirm that the script runs correctly and that the output displays as you expect it to.

Accessing a tuple by position

1 Like a list, the items of a tuple can also be accessed by position: 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

27. To access the `apple` string, enter the following code:

```
print(myFinalAnswerTuple[0])
```

28. To access the `banana` string, enter the following code:

```
print(myFinalAnswerTuple[1])
```

29. To access the `pineapple` string, enter the following code:

```
print(myFinalAnswerTuple[2])
```

30. Save and run the file.

31. Near the top of the IDE window, choose the **Run** (Play) button.

32. Confirm that the script runs correctly and that the output displays as you expect it to.

```
['apple', 'banana', 'cherry']  
<class 'list'>  
apple  
banana  
cherry  
['apple', 'banana', 'orange']  
('apple', 'banana', 'pineapple')  
<class 'tuple'>  
apple  
banana  
pineapple
```

Exercise 3: Introducing the dictionary data type

Defining a dictionary

A dictionary is a list with named positions (keys). Imagine that your list shows people's favorite fruit.

33. Return to the Python script, and enter the following code:

```
myFavoriteFruitDictionary = {  
    "Akua" : "apple",  
    "Saanvi" : "banana",  
    "Paulo" : "pineapple"  
}
```

34. Use the `print()` function to write the dictionary to the shell:

```
print(myFavoriteFruitDictionary)
```

35. Use the `type()` function to write the data type to the shell:

```
print(type(myFavoriteFruitDictionary))
```

36. Save and run the file.

37. Confirm that the script runs correctly and that the output displays as you expect it to.

Accessing a dictionary by name

In this activity, you will use the name of the individuals to get their favorite fruit, instead of numbers.

38. To access Akua's favorite fruit, enter the following code:

```
print(myFavoriteFruitDictionary["Akua"])
```

39. To access Saanvi's favorite fruit, enter the following code:

```
print(myFavoriteFruitDictionary["Saanvi"])
```

40. To access Paulo's favorite fruit, enter the following code:

```
print(myFavoriteFruitDictionary["Paulo"])
```

41. Save and run the file.

42. Confirm that the script runs correctly and that the output displays as you expect it to.

```

['apple', 'banana', 'cherry']
<class 'list'>
apple
banana
cherry
['apple', 'banana', 'orange']
('apple', 'banana', 'pineapple')
<class 'tuple'>
apple
banana
pineapple
{'Akua': 'apple', 'Saanvi': 'banana', 'Paulo': 'pineapple'}
<class 'dict'>
apple
banana
pineapple

```

Congratulations! You have worked with the list, tuple, and dictionary data types in Python.

End Lab

🚩 Congratulations! You have completed the lab.

43. Choose **End Lab** at the top of this page, and then select Yes to confirm that you want to end the lab.

A panel indicates that *DELETE has been initiated... You may close this message box now.*

44. A message *Ended AWS Lab Successfully* is briefly displayed, indicating that the lab has ended.

Additional Resources

For more information about AWS Training and Certification, see <https://aws.amazon.com/training/> (<https://aws.amazon.com/training/>).

Your feedback is welcome and appreciated. If you would like to share any suggestions or corrections, please provide the details in our AWS Training and Certification Contact Form (<https://support.aws.amazon.com/#/contacts/aws-training>).

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